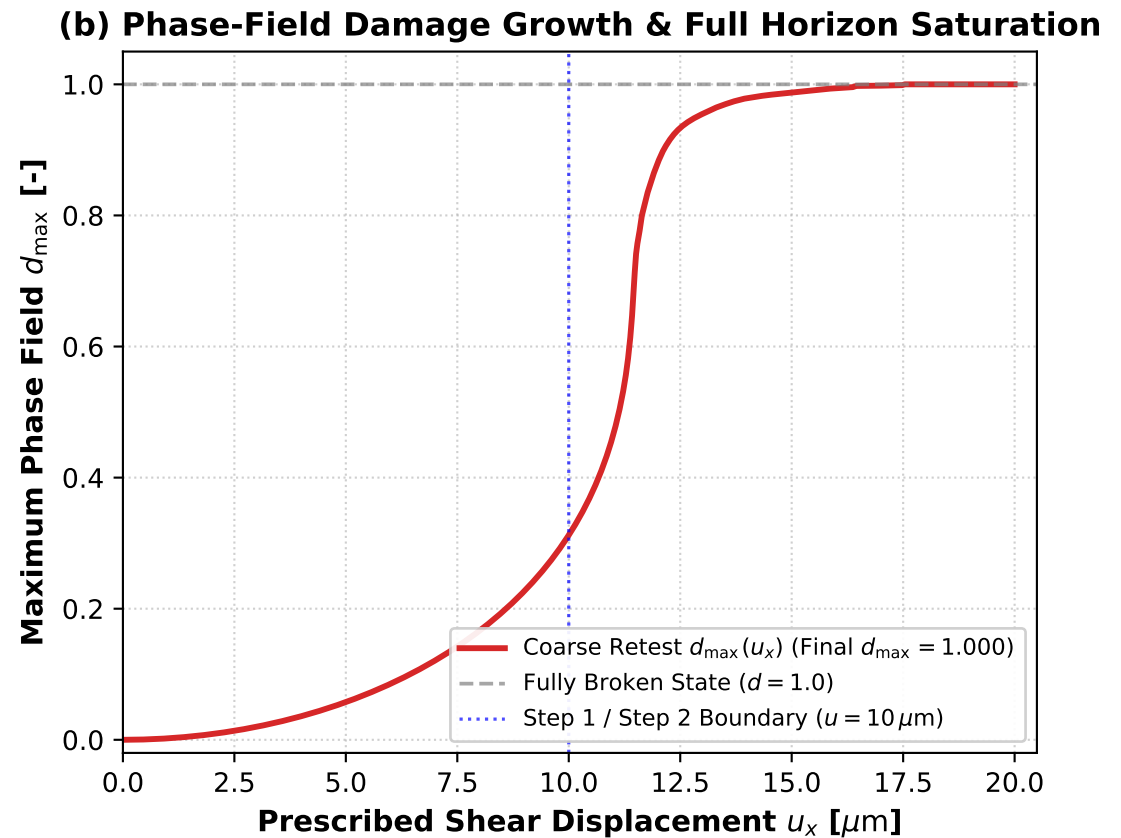
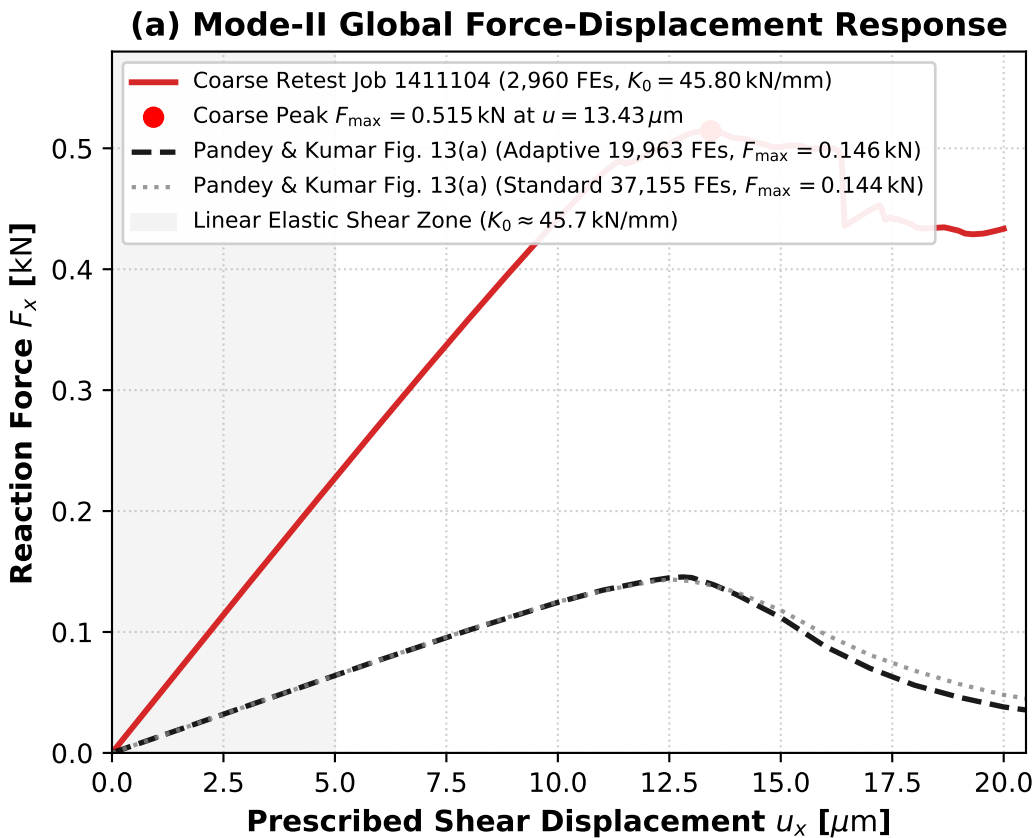
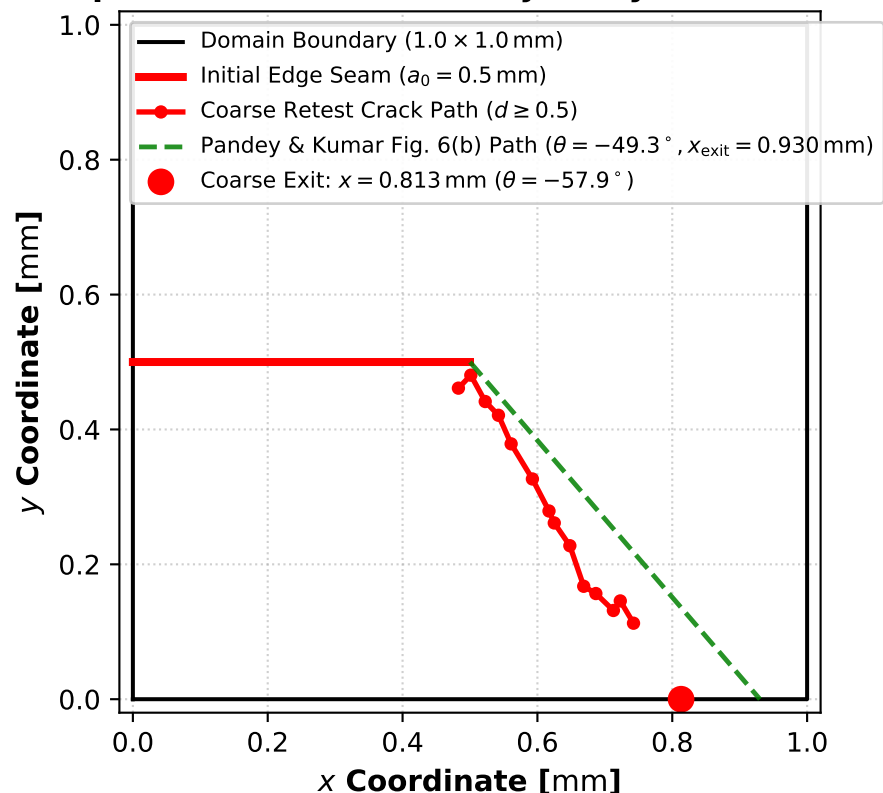




(c) Computed Mode-II Crack Trajectory vs Literature Path



(d) Gate M2-4 Discretization Metrics & Formulation Verification

Quantity / Metric	Coarse Retest (J1)	Adapted Retest (J2)	Literature Ref (Fig 13)
Finite Element Count	2,960 FEs	22,530 FEs	19,963 FEs (Adaptive)
Node Count	3,037 Nodes	22,642 Nodes	N/A (Published)
Initial Stiffness K_0	45.80 kN/mm	45.49 kN/mm (In-Situ)	12.80 kN/mm (Apparent)
Reaction Force at 4.15	189.9 N	188.58 N (In-Situ)	~ 53.0 N
Peak Load F_{max}	514.51 N	Solving (Active)	145.5 N (19.9k FEs)
Displacement $u(F_{\text{max}})$	13.43 μm	Solving (Active)	12.80 μm
Final Reaction Force	433.47 N (at 20 μm)	Solving (Active)	38.0 N (at 20 μm)
Softening Load Drop	15.75%	Solving (Active)	73.88%
Terminal Damage d_{max}	1.000000 (Full)	0.0528 (at 3.87 μm)	1.000 (Broken)
Crack Chord Angle	-57.95 deg	Active (Solving)	-49.30 deg
Bottom Exit ($y=0$)	$x = 0.813$ mm	Active (Solving)	$x = 0.930$ mm
Physical Solver Status	COMPLETED (Exit 0)	RUNNING (0 Cutbacks)	Published Baseline